

Servers on Fire:



Keeping OpenStreetMap Online

OSM Operations Team

State of the Map EU 2025

Grant Slater - en.osm.town/@Firefishy - 14 November 2025



**OSM.org should
just work**

100% of the time



Ops Team?



246 TB/month

planet.osm.org bandwidth usage - October 2025



1.4 PB/month

tile.osm.org bandwidth usage - October 2025



122 Billion Requests/month

tile.osm.org tile usage - October 2025



1 Billion Requests

OpenStreetMap.org - October 2025



Equinix Amsterdam

Server	Description	System	Distro	Stats	Last Contact
dribble	Taginfo server	HPE ProLiant DL360 Gen10	Ubuntu 22.04	prometheus	01:17
dulcy	Nominatim geocoding server	Supermicro SYS-1028U-TN10RT+	Ubuntu 22.04	prometheus	00:57
faffy	Development and “tool” server	HPE ProLiant DL360 Gen10	Debian 12	prometheus	01:27
ironbelly	Site gateway	Supermicro X9DR3-F	Debian 12	prometheus	01:20
karm	Read only database mirror for www.openstreetmap.org	Supermicro SYS-2028U-TN24R4T+	Ubuntu 22.04	prometheus	01:05
norbert	Backup server	Supermicro SYS-120U-TNR	Ubuntu 22.04	prometheus	01:17
	Web server for planet.openstreetmap.org Planet file generation server				
odin	Tile server	Supermicro SYS-1029P-WTRT	Ubuntu 22.04	prometheus	01:11
snap-01	Primary database server for www.openstreetmap.org	Supermicro SYS-2029U-TN24R4T	Ubuntu 22.04	prometheus	01:11
spike-06	Frontend web server for www.openstreetmap.org	HP ProLiant DL360 Gen9	Debian 12	prometheus	01:22
spike-07	Frontend web server for www.openstreetmap.org	HP ProLiant DL360 Gen9	Debian 12	prometheus	01:21
spike-08	Frontend web server for www.openstreetmap.org	HP ProLiant DL360 Gen9	Debian 12	prometheus	01:19
vhagar	Nominatim geocoding server	HPE ProLiant DL360 Gen10	Ubuntu 22.04	prometheus	00:56

Equinix Dublin

Server	Description	System	Distro	Stats	Last Contact
culebre	Tile server	Supermicro AS -1124US-TNRP	Ubuntu 22.04	prometheus	01:03
fafnir	Site gateway	HP ProLiant DL360 Gen9	Ubuntu 22.04	prometheus	01:07
	Mail server				
fume		HPE ProLiant DL360 Gen10	Debian 12	prometheus	01:18
gorwen	Discourse server	HP ProLiant DL360e Gen8	Ubuntu 22.04	prometheus	01:02
grisu	Overpass server	HPE ProLiant DL360 Gen10	Debian 12	prometheus	01:04
horntail	Web server for planet.openstreetmap.org	Supermicro SYS-120U-TNR	Ubuntu 22.04	prometheus	01:04
idris	Chef server	HP ProLiant DL360 Gen9	Debian 12	prometheus	01:02
	DNS management server				
	Git server				
	Let's Encrypt certificate management server Network configuration backup server				
konqi	Wiki server	HP ProLiant DL360 Gen9	Ubuntu 22.04	prometheus	01:08
longma	Nominatim geocoding server	Supermicro AS -1124US-TNRP	Ubuntu 22.04	prometheus	01:05
muidris	GPS tile server	HPE ProLiant DL360 Gen10	Ubuntu 22.04	prometheus	01:01
naga	Subversion redirector	HP ProLiant DL360 Gen9	Debian 12	prometheus	01:19
	Trac redirector				
	IRC redirector				
	Blog aggregator				
osun	Mail server for web site statistics	HPE ProLiant DL360 Gen10	Ubuntu 22.04	prometheus	01:23
shin	Planet database mirror for www.openstreetmap.org	Supermicro SYS-120U-TNR	Ubuntu 22.04	prometheus	01:17
spike-01	Frontend web server for www.openstreetmap.org	HP ProLiant DL360 Gen9	Debian 12	prometheus	01:22
spike-02	Frontend web server for www.openstreetmap.org	HP ProLiant DL360 Gen9	Debian 12	prometheus	01:19
spike-03	Frontend web server for www.openstreetmap.org	HP ProLiant DL360 Gen9	Debian 12	prometheus	01:09

University College London

Server	Description	System	Distro	Stats	Last Contact
eddie	Read only database mirror for www.openstreetmap.org	Supermicro SYS-2028U-TN24R4T+	Ubuntu 22.04	prometheus	01:21
ridley	Site gateway	HP ProLiant DL360 G6	Ubuntu 20.04	prometheus	00:52
	Foundation related sites				
snap-02	Read only database mirror for www.openstreetmap.org	Supermicro SYS-2029U-TN24R4T	Ubuntu 22.04	prometheus	01:24
ysera	Tile server	Supermicro SYS-1029P-WTRT	Ubuntu 22.04	prometheus	01:18

hardware.osm.org

- Home
- Starred
- Dashboards
- Alerting
- Administration

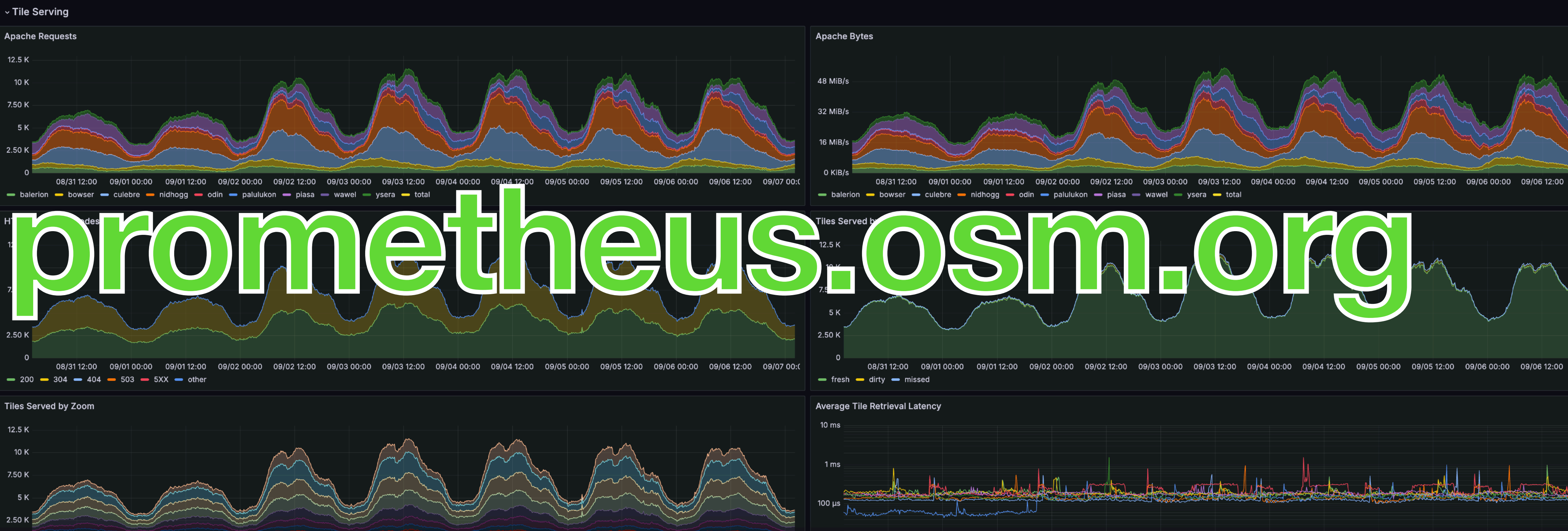
Host

All



Service Summary

Host ↑	Replication Delay	Queue Length	Dropped	Rendered	Fresh Tiles	Dirty Tiles	Missed Tiles
balerion	1.60 mins	0	0 tps	6.62 tps	538 tps	4.33 tps	
browser	48.3 s	0	0 tps	6.52 tps	415 tps	6.08 tps	
culebre	1.74 mins	0	0 tps	4.19 tps	541 tps	1.03 tps	
nidhogg	1.78 mins	0	0 tps	4.08 tps	531 tps	0.971 tps	
odin	1.72 mins	0	0 tps	1.91 tps	128 tps	1.18 tps	
palulukon	1.81 mins	0	0 tps	9.14 tps	689 tps	5.59 tps	
piasa	1.79 mins	0	0 tps	8.23 tps	623 tps	4.14 tps	
wawel	1.70 mins	0	0 tps	0 tps	0 tps	0 tps	





Drive Bay ID:
1 3 5 UMB
2 4 6 7 8

VHAGAR

DULCY

SPIKE-08

SPIKE-07

SPIKE-06

ProLiant
DL360
Gen10

hp
ProLiant
DL360
Gen9

hp
ProLiant
DL360
Gen9

hp
ProLiant
DL360
Gen9

5x Incidents

Incident #1

What does this do?

 block.js

Raw

```
1  function checkIsOnline() {
2      setInterval(function () {
3          var tm = new Date().toISOString();
4          $http.get('https://api.openstreetmap.org/api/0.6/notes/search?q=abcde&t=' + tm).then(function (res) {
5              //console.log("ApplicationOnlineIndicator running");
6              eventDispatcher.fireEvent("internet_connection_changed", true);
7          }, function (err) {
8              // console.log("ApplicationOnlineIndicator running ", err);
9              eventDispatcher.fireEvent("internet_connection_changed", false);
10         });
11     }, 3000);
12 }
13 checkIsOnline();
```

Incident #1 (2020): CCTV app's “online check”

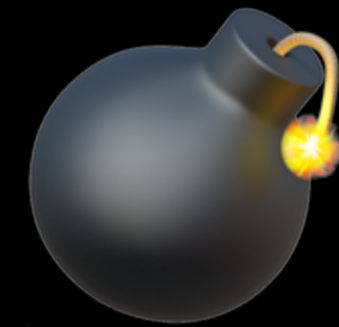
 block.js

Raw

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1  function checkIsOnline() {
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11     }, 3000);
12 }
13 checkIsOnline();
```

Incident #1 (2020): CCTV app's “online check”

Audience Poll



Access Denied

“Zip Bomb”



Incident #2

What do they actually do?

Perfect Solution for Apps & Games Monetization

A peer-to-business ecosystem connecting user-driven software monetization with companies

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Increase your revenue without ads and in-app purchases

Embed a
hidden
web proxy
server in
apps to allow
AI scrapers
to bypass
website filters

Perfect Solution for Apps & Games Monetization

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• WEB DATA MANAGEMENT • SCRAPER APIS • INFATICA SDK

A new ethical way to monetize your apps

Increase your revenue without ads and in-app purchases

Incident #2 (2024 ongoing):

Residential proxy-SDK scraping

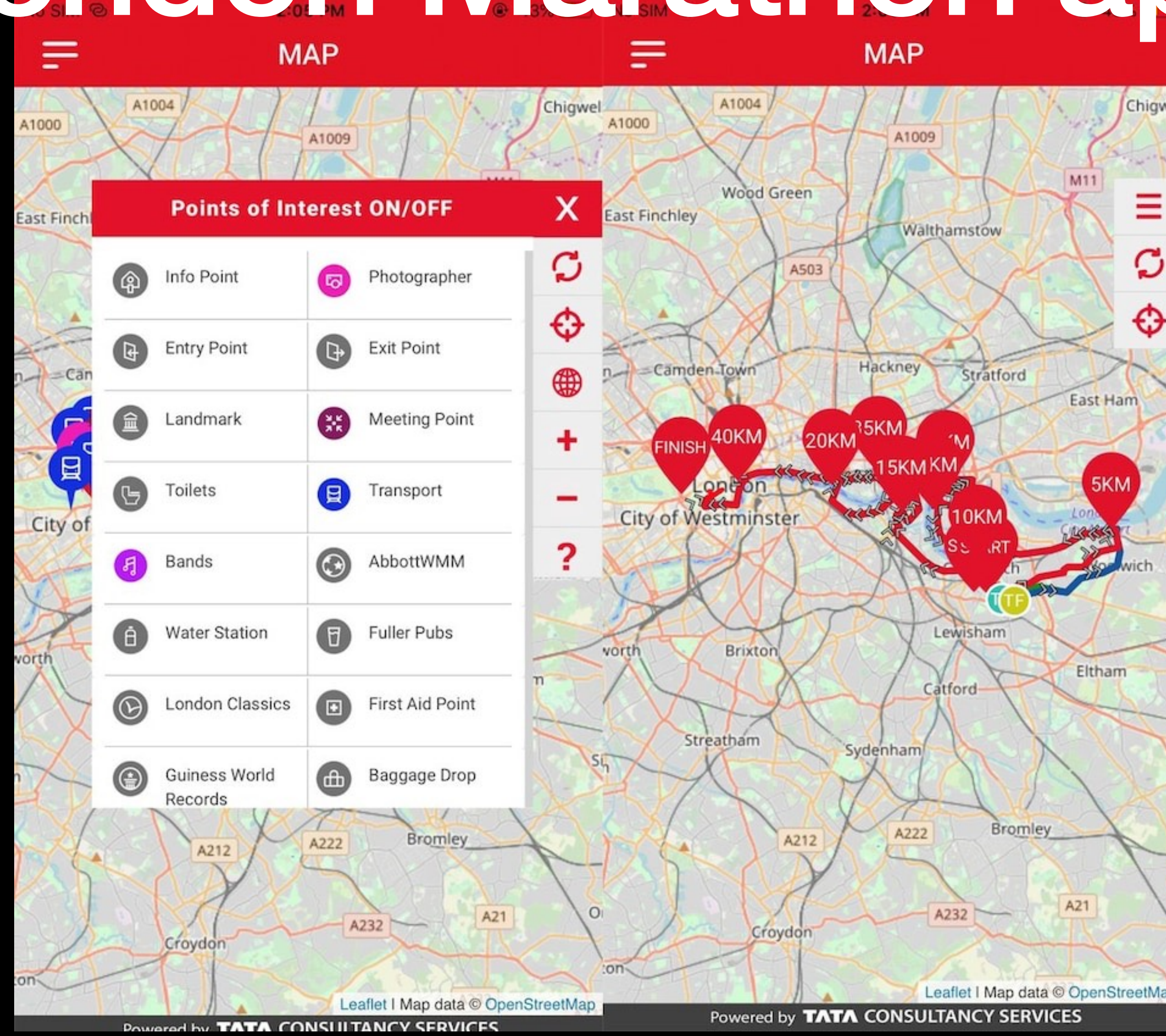
- Ongoing battle to detect
- Rate limits
- Deploy Anubis for are you a Human detection?
- Monitoring / Alerting / Blocks

Incident #3

Why was this an issue?

Official 2019 London Marathon app

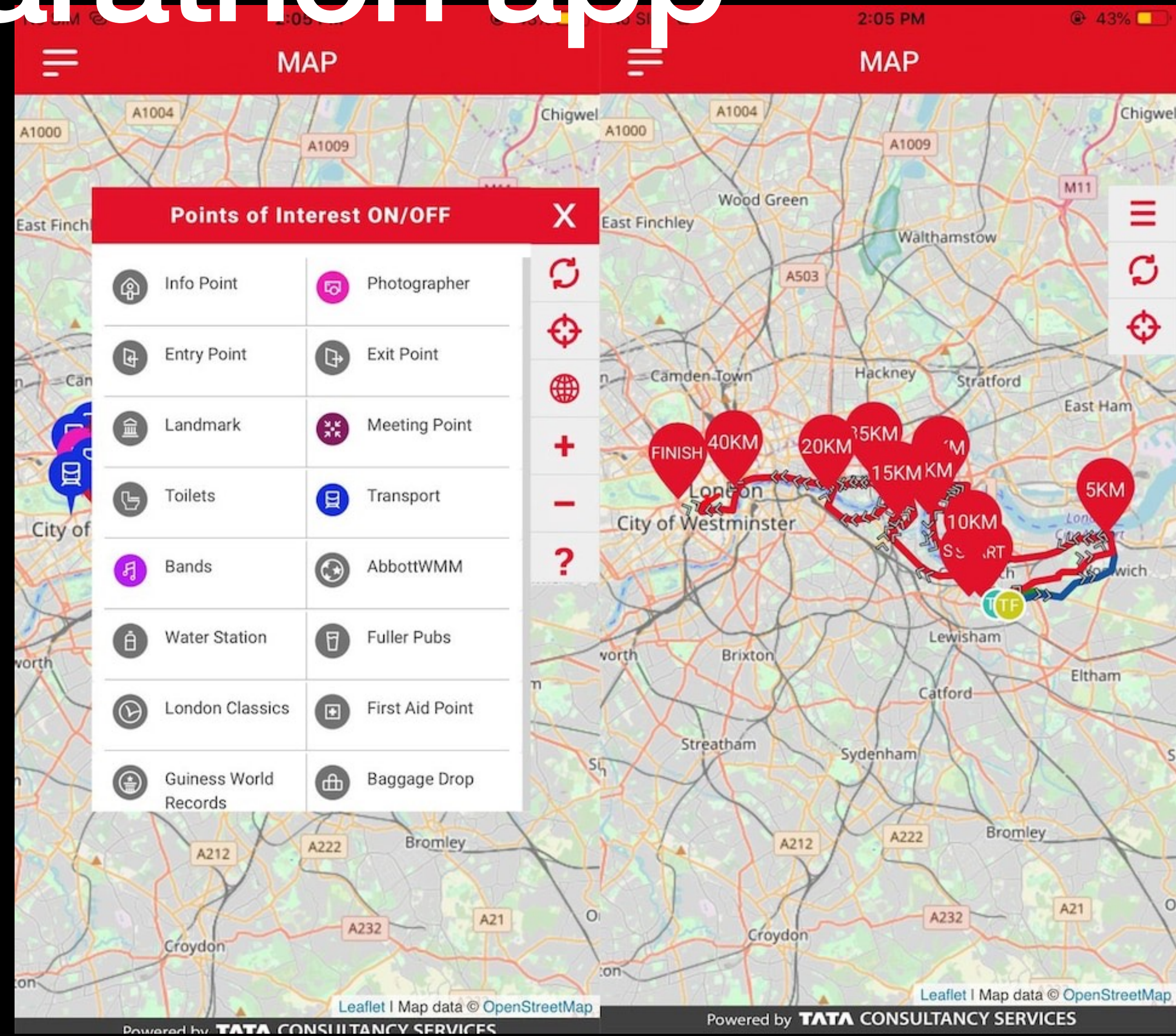
Incident #3 (2019): London Marathon app



Incident #3 (2019): London Marathon app

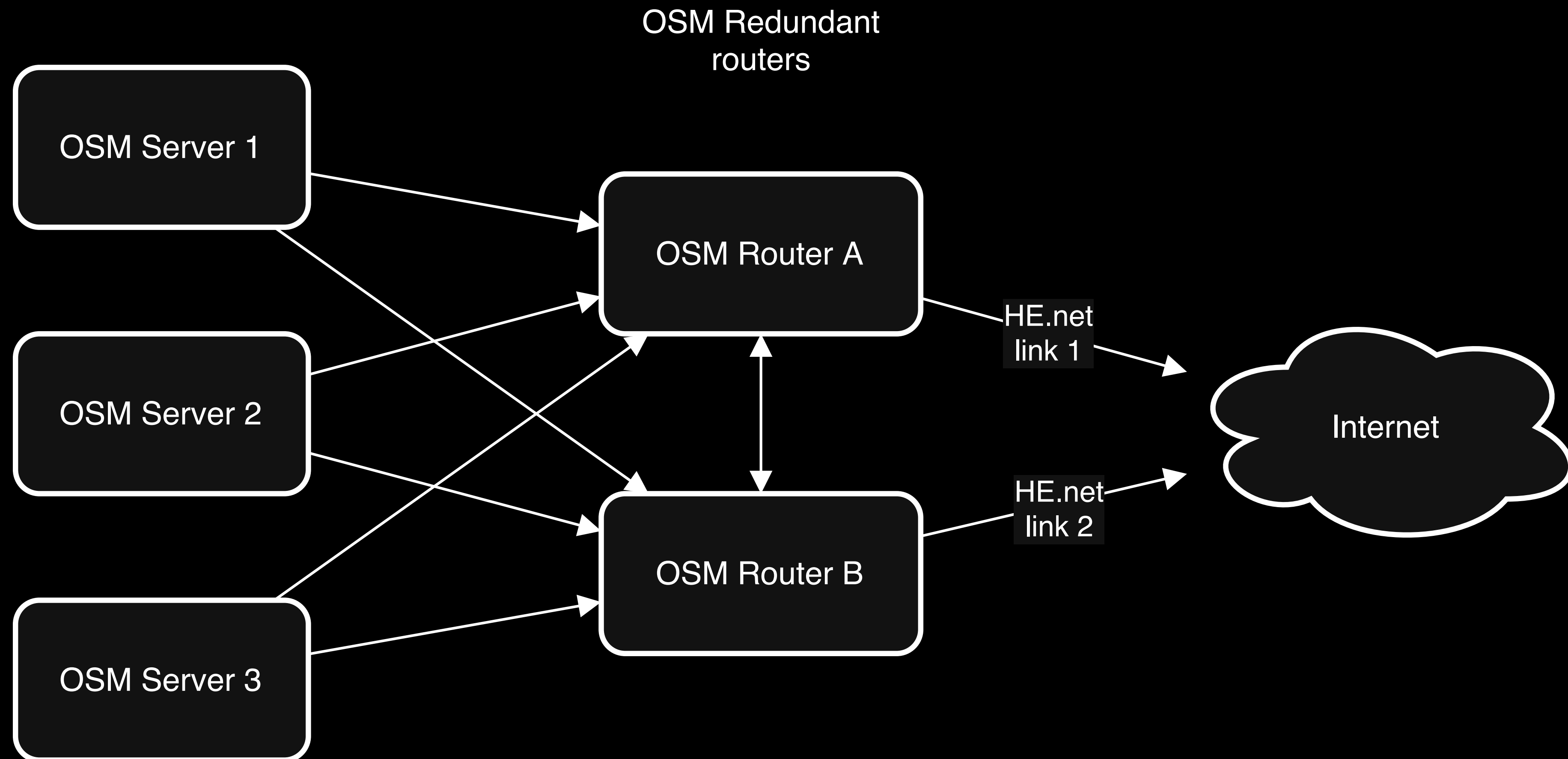
- 500,000+ app downloads
- We hadn't been told, discovered on the day.
- Servers 🔥🔥🔥 maxing out network

Mitigated by us manually redistributing the traffic across our global server pool for 12 hours



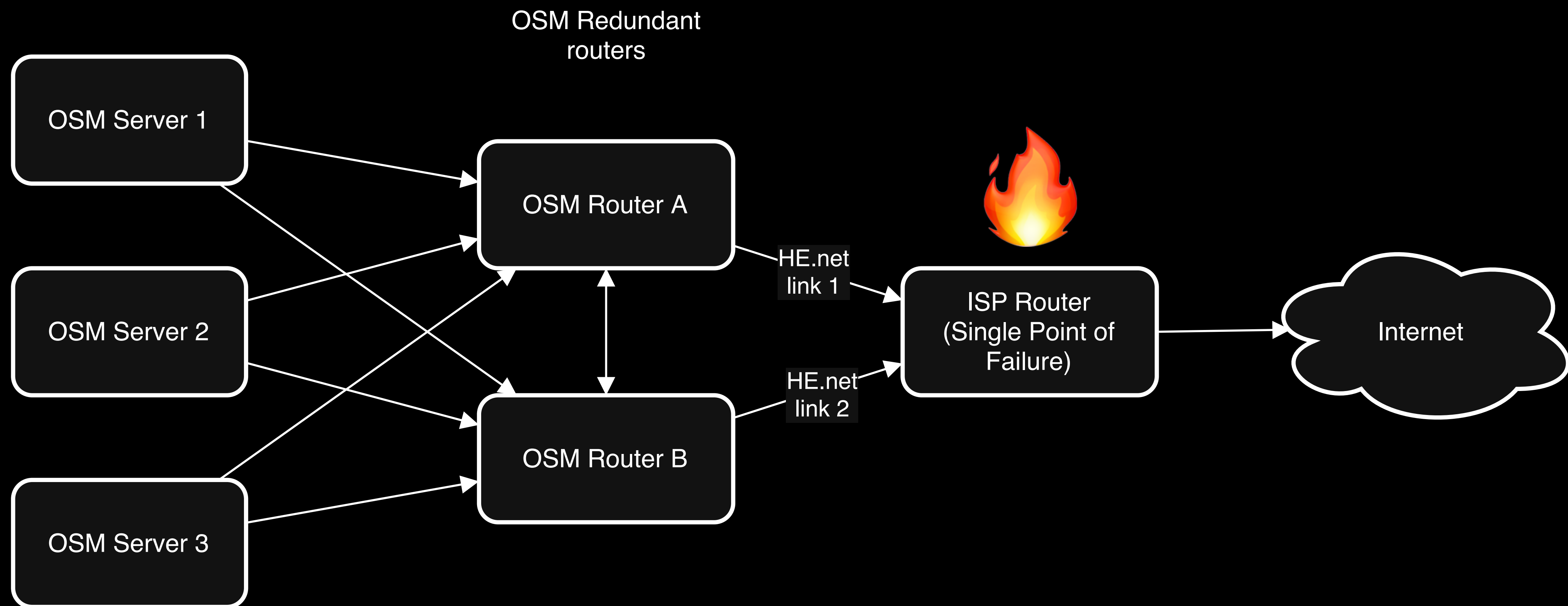
Incident #4

Why was this an issue?



Incident #4

Why was this an issue?

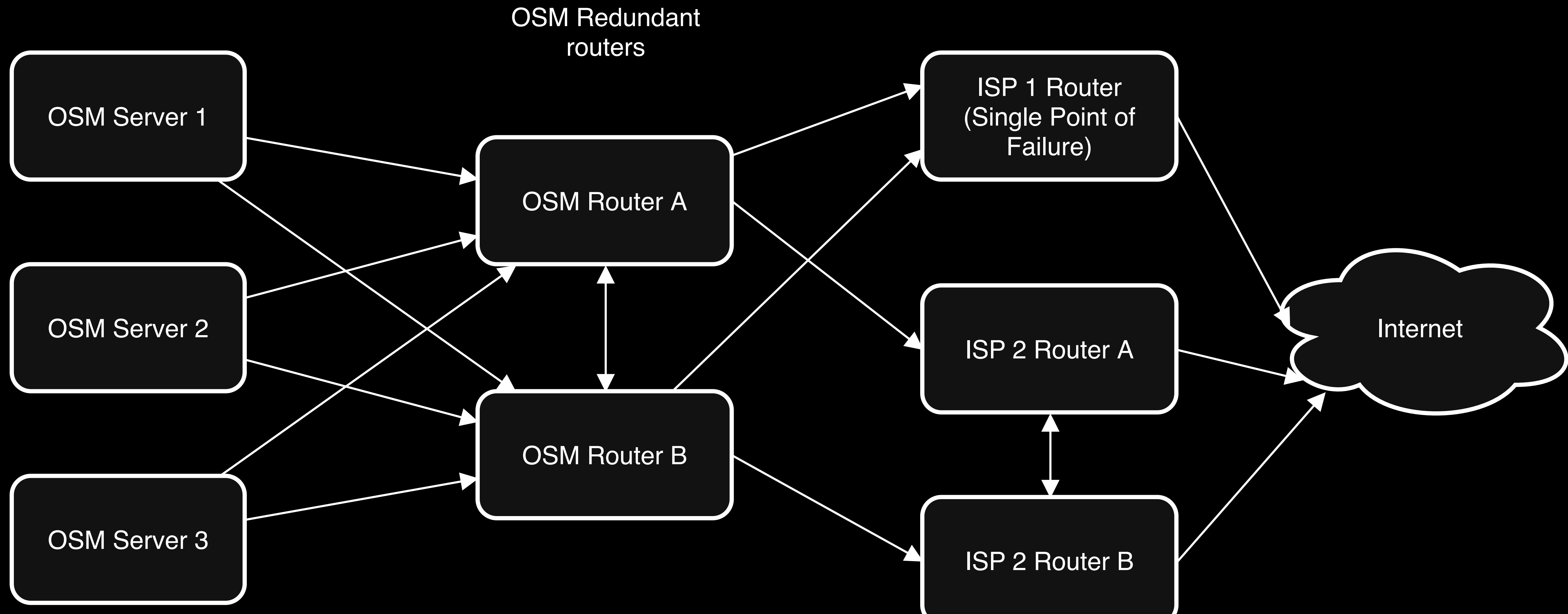


Incident #4 (2024):

ISP side equipment failure - no redundancy

- ISP had no backup hardware in Europe. Replacement had to be shipped from California
- We'd be misled about redundancy
- We switched to another provider during our ISP's outage
- 4G modem backup link critical
- Equinix Internet now provide a true redundant + resilient internet connection without a single point of failure

<https://operations.osmfoundation.org/2025/02/15/post-mortem.html>



Incident #5

Why was this an issue?

Email...

To: OSM Operations

Subject: *Research Request - Higher Bandwidth Limits for Changeset Analysis*

Incident #5

Why was this an issue?

To: OSM Operations

Subject: Research Request - Higher Bandwidth Limits for Changeset Analysis

... I've been downloading changesets via the API (/changeset/{id}/download) from [University] network, but after ~8 hours of downloading, I encounter HTTP 509 (Bandwidth Limit Exceeded) errors. At the current rate:

- Download speed: ~70 successful downloads/sec (with 20 parallel workers)*
- Error rate: 97% due to bandwidth throttling*
- Estimated completion time: 5-7 months [...]*

Incident #5 (ongoing):

Download *all the things* via mapping API

- OSM.org API is for Mappers
- Usage Policy: <https://operations.osmfoundation.org/policies/api/>
- All our map data is available via bulk download services (planet.osm.org)

OSM Operations Working Group

What do we do?

- Servers & Hardware Infrastructure
- Manage Software & setup
- Backups / Redundancy / Availability
- Scaling / Growth Planning
- Budgeting
- Service Usage Policies

OSM Operations Working Group

How do we communicate?

en.osm.town/@osm_tech

github.com/openstreetmap/operations/issues

irc.osm.org #osmf-operations / #osm-dev
(Discussion, Chef changes)

Private Email List
(Alerts, Failures, Planning, Approvals)

Online Video Call every 2nd week



OpenTofu[📦]



fastly[®]



EQUINIX



CHEF



debian



Any Questions?

operations.osmfoundation.org

en.osm.town/@osm_tech
en.osm.town/@Firefishy

Online Video call every 2nd week

Join? Email: [operations AT osmfoundation.org](mailto:operations@osmfoundation.org)



OpenStreetMap Ops Team

@osm_tech@en.osm.town

Do you have deep #AWS #SecOps or #DevOps experience? We're seeking help to harden #OpenStreetMap's AWS accounts 🗝️. Exploring IAM Roles Anywhere; most infra is #OpenTofu #IaC. Stack includes EC2 (limited), S3 + replication, Athena/Glue, CloudWatch & CloudTrail. Come chat to us at @sotmeu 2025.stateofthemap.eu/ #OSM #sotmeu #SotMEU2025



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